

**Project Yorozya: A Variable Reward-Based Serious Game Design for Supporting Task
Initiation and Sustained Engagement in Young Adults with ADHD**

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Introduction

Attention-deficit/hyperactivity disorder (ADHD) is a neurodevelopmental condition characterized by persistent deficits in attention regulation, inhibitory control, and executive functioning (American Psychiatric Association, 2013). These deficits extend well beyond the surface-level symptoms of inattention and impulsivity; in practice, they manifest as difficulties with task initiation, sustained effortful engagement, and the kind of goal-directed self-regulation that most routine tasks require (Barkley, 2012). For young adults in particular, these challenges carry significant consequences in academic and professional contexts, where demands for self-directed attention, independent task management, and sustained cognitive effort are pervasive and difficult to avoid.

The present study does not treat ADHD as a monolithic diagnostic category. Instead, it focuses on a specific functional problem that sits at the core of the condition: the difficulty of initiating and sustaining attention-directed behavior over time, especially in contexts characterized by low intrinsic reward or high cognitive load. This is also where executive dysfunction tends to be most apparent. Sustaining attention is not a passive process; it requires the concurrent engagement of working memory, self-monitoring, and effortful inhibition of competing responses (Diamond, 2013), all of which are reliably impaired in ADHD.

Intervention research in this space has increasingly turned toward cognitive training and game-based digital therapeutics as possible alternatives or supplements to pharmacological treatment (Cortese et al., 2015). Serious games, in particular, have been explored for their potential to create structured, feedback-rich environments that might be more intrinsically engaging than conventional training paradigms. The evidence to date is cautiously promising, but a consistent finding across reviews is that the effectiveness of these tools is heavily contingent on design quality, and that engagement tends to drop off sharply with repeated use (Sújar et al., 2022; Rodríguez Timaná et al., 2024). This is not a minor limitation. For a population already prone to disengagement, an intervention that fails to sustain participation is unlikely to produce meaningful outcomes regardless of its theoretical basis.

The present study takes that problem as its starting point. The goal is not to develop a comprehensive ADHD intervention, but to address a more tractable design question: how might a serious game be structured to support focus initiation, task persistence, and continued engagement in individuals with ADHD? The core challenge this study responds to is the overlap between two reinforcing difficulties, namely the attentional and executive-function impairments that define the condition, and the tendency of existing support tools to lose users before any lasting effect can take hold.

Literature Review

Epidemiological data indicate that approximately 6.0% of adults in the United States currently carry an ADHD diagnosis, equivalent to an estimated 15.5 million individuals; prevalence is highest among those aged 18 to 24, at 21.7% (Staley et al., 2024). Although ADHD is frequently conceptualized as a childhood condition, substantial evidence confirms that its functional impairments persist into adulthood and often become more consequential as academic and professional demands increase. The functional burden of the condition is not uniformly distributed across symptom domains: a significant proportion of adults with ADHD demonstrate measurable impairments in executive functioning, with documented deficits in sustained attention, inhibitory control, and working memory that are associated with poorer academic and occupational outcomes (Biederman et al., 2006). Taken together, these patterns suggest that the primary difficulty in ADHD is less a matter of diagnostic classification than one of functional regulation: specifically, the persistent challenge of maintaining goal-directed behavior across time and context.

Within this framework, sustained attention warrants particular attention as both a theoretically central construct and a clinically relevant intervention target. Sustained attention is not a unitary or passive cognitive state; rather, it reflects the coordinated operation of multiple executive processes, including working memory updating, self-monitoring, response inhibition, and the effortful regulation of arousal and motivation (Barkley, 2012; Posner & Petersen, 1990). In individuals with ADHD, the breakdown of this coordination is reliably exacerbated under conditions of delayed reward, low task novelty, or extended cognitive demand, precisely the conditions that characterize academic work and independent task completion (Sergeant, 2005). This pattern has led some researchers to reconceptualize ADHD not primarily as an attentional disorder but as a disorder of effort regulation and motivational sensitivity (Volkow et al., 2009), a framing with direct implications for how interventions should be designed.

Understanding ADHD through the lens of executive dysfunction and effort regulation also helps explain a recurring limitation in the intervention literature: the difficulty of sustaining user engagement over time. If the core impairment involves sensitivity to reward delay and reduced intrinsic motivation for effortful tasks, then interventions that rely on sustained voluntary participation face a structural challenge that goes beyond content quality or task design. This observation provides the theoretical foundation for the present study's focus on engagement architecture as a design problem in its own right.

Stakeholder Interview / User Research

To complement the theoretical grounding established through the literature review, this study incorporates qualitative stakeholder interviews with individuals diagnosed with ADHD. This methodological choice reflects a commitment to grounding design decisions in lived experience rather than in theoretical inference alone. While existing research provides substantial evidence regarding the cognitive and neurological characteristics of ADHD, the experiential dimensions of attentional difficulty, particularly in relation to tool use, disengagement, and motivational sensitivity, cannot be fully resolved through literature alone. The interviews are therefore not treated as supplementary to the design process, but as a constitutive part of it, serving as an early needs assessment that informs both the conceptual framing of the problem and the specific mechanics of the proposed intervention.

The interview protocol is organized around three lines of inquiry. The first examines participants' lived experiences of attention-related difficulties in ecologically valid contexts, specifically the conditions under which task initiation fails, sustained effort diminishes, or prolonged cognitive persistence becomes sufficiently effortful to prompt disengagement. Particular attention is given to academic and professional environments, where demands for self-directed and sustained cognitive engagement are typically highest. The second line of inquiry focuses on participants' prior experiences with productivity tools, digital interventions, and game-based systems, with the aim of identifying not only which features were experienced as supportive, but also the circumstances under which engagement deteriorated or ceased altogether. The third explores participants' responses to specific game design elements, including reward structures, progression systems, collection-based mechanics, and narrative framing, in order to distinguish mechanics perceived as motivating from those experienced as distracting or cognitively burdensome.

Taken together, these three dimensions are intended to surface patterns that quantitative or experimental methods are ill-suited to capture: the subjective thresholds at which engagement breaks down, and the design features that users identify retrospectively as unsustainable. The findings generated through this stage directly inform the design decisions documented in the subsequent sections of this study.

Problem Statement

The central problem addressed in this study sits at the intersection of two related but distinct challenges. The first is clinical and cognitive in nature: individuals with ADHD demonstrate reliable impairments not only in sustained attention but in the broader constellation of executive processes that support goal-directed behavior, including task initiation, effort maintenance, and

persistence under conditions of low reward or high cognitive demand. The second is an intervention design problem: despite growing interest in digital therapeutics and game-based tools as ADHD support mechanisms, existing approaches consistently struggle to maintain user engagement beyond initial exposure (Lin & Chang, 2025). The structural demands of repetitive training paradigms are particularly poorly matched to the motivational profile of users with ADHD, for whom disengagement is not a failure of willpower but a predictable consequence of the condition itself.

Stakeholder interviews conducted as part of this study elaborate the experiential texture of this problem. Participants described breakdown in focus as rarely sudden; rather, disengagement typically followed a recognizable sequence involving stalled task initiation, diminishing feedback salience, and the absence of visible progression markers that might otherwise sustain motivation. These accounts are consistent with neuropsychological accounts of ADHD as a disorder of delayed reward sensitivity and effort regulation, and they underscore a limitation that the empirical literature has not fully resolved: the gap between interventions that are theoretically sound and those that users will actually sustain.

The scope of this problem is intentionally bounded. This study does not aim to address ADHD as a comprehensive clinical condition, nor does it seek to intervene across all symptom domains. The focus is placed specifically on the behavioral and experiential challenges surrounding focus initiation, task persistence, and continued participation in support systems. Understood in these terms, the problem is neither purely clinical nor purely technical, but located in the design of engagement itself: specifically, in the persistent mismatch between the attentional and motivational characteristics of individuals with ADHD and the structural logic of existing tools, which tend to rely on repetitive, low-salience interaction patterns that are poorly suited to sustaining participation in this population.

The question this study pursues is therefore the following: how might a serious game be structured to support focus initiation, task persistence, and longitudinal engagement for individuals with ADHD, not by circumventing the attentional difficulties associated with the condition, but by building intervention logic around them?

Design Rationale and System Description

The design of this intervention is organized around a single organizing principle: that engagement architecture must be built around the attentional and motivational profile of individuals with ADHD, rather than treating engagement as a secondary concern to be addressed after core functionality is established. This principle generates three interconnected design priorities. First, the system must lower the threshold for task initiation, since focus initiation is often the point at which behavioral breakdown first occurs in individuals with ADHD. Second, it must provide immediate, salient feedback at intervals short enough to sustain motivation across a

session. Third, it must offer long-term accumulation mechanics that give users a reason to return across days and weeks, without relying on the kind of intrinsic motivation for repetitive tasks that individuals with ADHD reliably find difficult to sustain.

To meet these requirements, the intervention takes the form of a serious game built around a light narrative framing: the player assumes the role of the owner of a one-person yorozuya, a small general-services shop, in a fictional town. This framing is not incidental to the design. Narrative context has been shown to increase intrinsic motivation and task salience in game-based interventions by providing a coherent identity and purpose for user actions (Sújar et al., 2022). By situating focus sessions within the logic of "opening the shop" and completing "requests" from town residents, the system reframes the act of beginning work as a meaningful in-world event rather than a neutral behavioral trigger. This diegetic framing connects otherwise abstract actions to a role-based context, aiming to reduce the psychological friction associated with task initiation in ADHD without adding mechanical complexity.

Design Implementation

The prototype was developed in Unity and is intended for young adults with ADHD who experience difficulty sustaining attention across academic and everyday tasks. The implementation translates the engagement-centered design principles outlined in the preceding section into a functional system by operationalizing three interlocking layers: a core focus and task management layer, a feedback and reward layer, and a long-term retention layer. Given the time constraints of the current project phase, the current build constitutes a minimum viable product in which the core focus timer, journal system, and primary reward mechanics are fully functional. Several features within the feedback and reward layer and the long-term retention layer, including extended customization options, the full collectible item pool, and more complex progression structures, remain partially implemented. These elements are included in the design documentation to illustrate the intended engagement architecture of the complete system and its potential for further development in subsequent iterations.

The current build presents users with a single-screen interface in which the focus timer occupies the visual and functional center of the display. This layout is deliberate: by anchoring the interface around the timer rather than around reward or progression elements, the design preserves the primacy of the productivity function and reduces the risk that gamified elements will distract from or displace the core task. All additional systems are accessible from the periphery of this central view and are designed to be engaged during natural breaks rather than during active focus sessions.

Core Focus and Task Management Layer

At the start of each session, users activate the focus timer by "opening the shop," a narrative action that signals the beginning of a work period within the diegetic framing of the system. The timer supports configurable work intervals and break periods, functioning similarly to existing focus tools while remaining embedded within the game's narrative context. On first launch, users are prompted to define a daily working hour goal, which is displayed as a continuous progress indicator throughout the day. This visibility is intentional: making accumulated focus time legible in real time allows users to perceive incremental progress even within sessions that do not reach the full daily target.

The journal system is accessible from the main interface and functions as the primary task management tool. Users enter tasks as named "requests," each of which can optionally carry a deadline. The interface displays active requests alongside the timer, allowing users to reference their task list without navigating away from the focus session. Incomplete tasks persist into subsequent days automatically, requiring no manual carry-over action from the user. This persistence is implemented as a default rather than an opt-in feature, reflecting the design decision to remove rather than defer the administrative burden associated with task management in individuals with ADHD.

Feedback and Reward Layer

Two in-game currencies are displayed in the upper-left corner of the interface and remain visible throughout each session. Coins are earned upon the completion of focus sessions and function as the primary currency for direct item purchases. Stamps are awarded through a combination of session completion, daily goal attainment, and daily login, and are allocated to the mail system. By keeping both currencies visible throughout the session, the interface maintains a continuous passive reminder of the reward value associated with sustained engagement, without requiring users to navigate to a separate rewards screen.

The daily activity record is accessible from the lower-right section of the interface alongside the shop, mail system, and item storage. Upon meeting defined daily milestones, the activity record distributes stamps automatically. The mail system accepts stamps and returns randomized cosmetic items drawn from a pool of furniture, decorations, and in-game objects. In the current build, the item pool is populated with an initial set of collectibles sufficient to demonstrate the variable reward mechanic; the full range of collectible content remains under development.

Long-Term Retention Layer

Items obtained through the mail system and the in-game shop are used to furnish and decorate a persistent virtual space representing the player's yorozuya. In the current prototype, this space is rendered as a navigable environment that updates visually as new items are placed. The

accumulation of items over time produces a space that is legible as a record of the user's engagement history, with a more densely furnished environment reflecting more sustained participation. This visual progression is designed to function independently of numerical performance metrics, providing a form of longitudinal feedback that does not penalize inconsistent participation.

The secondary currency system, through which coins earned from completed journal tasks can be spent in a direct-purchase shop, is implemented in the current build with a limited initial inventory. This system operates in parallel with the randomized mail mechanic and is intended to offer users a predictable, controllable progression pathway alongside the variable reward structure. The music box feature, which allows background audio to be played during focus sessions, is implemented as a functional element in the current prototype, though integration with external audio sources remains a planned extension beyond the current build.

User Testing and Preliminary Findings

To evaluate the motivational and engagement potential of the proposed system, preliminary user testing was conducted with two participants clinically diagnosed with ADHD. Participants were recruited through convenience sampling. All participants held a formal clinical diagnosis of ADHD confirmed by a medical or psychiatric professional, and provided informed consent prior to testing. Given the small sample size and the early-stage nature of the prototype, the findings reported here are treated as exploratory rather than evaluative, intended to surface initial patterns in engagement behavior and identify design directions for future iteration rather than to establish generalizability.

Due to the incomplete implementation of the internal reward system at the time of testing, an external reward structure was adopted to approximate the intended feedback loop. Participants self-reported the number of stamps earned through completing focus sessions, which were then converted into equivalent reward pulls within a gacha-based system of their own choosing. This substitution was designed to isolate the motivational effect of effort-contingent reward delivery under conditions of high reward salience, without requiring the internal system to be fully operational. It is acknowledged that this approach introduces a confound: the engagement observed during testing may reflect participants' pre-existing investment in the external reward system rather than the prototype's standalone motivational architecture. Findings are interpreted with this limitation in mind.

Participant 1

The first participant, a recent university graduate, described their attentional patterns as highly variable and contingent on personal interest. Tasks that did not immediately engage their

attention were difficult to initiate and sustain, whereas intrinsically motivating activities could support extended periods of uninterrupted focus. This profile is consistent with interest-dependent attention commonly documented in ADHD (Barkley, 2012).

During testing, the participant reported a marked increase in motivation upon recognizing that task completion would translate directly into rewards within a preferred gacha system. However, they also noted that this effect may have been partially inflated by their existing emotional investment in the external system, and expressed uncertainty about whether equivalent engagement would be sustained if rewards were instead derived from the prototype's internal mechanics. This self-reported qualification is itself a meaningful finding: it suggests that reward salience is a critical moderating variable in the effectiveness of effort-contingent feedback systems for this population, and that the perceived value of rewards relative to user preference may matter as much as the structural properties of the reward mechanism itself.

Participant 2

The second participant, a graduate student in education, reported a similar interest-dependent attentional profile and emphasized the role of environmental structure in supporting focus. Distraction was more likely in socially engaging contexts, whereas self-directed and structured settings were more conducive to sustained task completion.

This participant did not regularly engage with gacha-based systems, and a substitute reward structure based on a customization-oriented game was used in place of the original external system. Initial testing produced increased motivation and self-reported productivity; however, this effect attenuated over time, consistent with a novelty effect in which initial engagement with a new system declines as familiarity increases (Fang et al., 2025). Despite this attenuation, the participant indicated that the system continued to support task initiation more reliably than the focus tools they typically used, suggesting that the structural benefits of the intervention may persist beyond the novelty phase even when motivational intensity diminishes.

The participant evaluated the journal and daily task structure positively, describing it as balanced and non-punitive in tone. However, they identified a limitation in the progression system: customization-based rewards alone did not generate a sufficiently strong sense of cumulative achievement over time. They suggested that more structured and visible forms of long-term progression could better sustain engagement across repeated use.

Cross-Case Analysis

Taken together, the two cases surface three preliminary patterns that carry implications for the next iteration of the design. First, reward salience appears to function as a critical moderating variable: when effort-contingent rewards were associated with systems participants found personally meaningful, engagement increased substantially. This finding is consistent with

neuropsychological accounts of ADHD as a disorder of motivational sensitivity rather than attention per se (Volkow et al., 2009), and suggests that the internal reward system of the prototype will need to achieve sufficient perceived value to replicate the engagement effects observed under external reward conditions.

Second, both participants exhibited patterns consistent with a novelty effect, in which initial engagement was relatively high but attenuated with repeated exposure. This finding reinforces the design rationale for a long-term retention layer: short-term motivational effects are unlikely to sustain participation over the timescales required for meaningful intervention, and progression systems must therefore provide reasons for re-engagement that extend beyond initial novelty.

Third, the cases suggest that different users may respond differently to variable versus deterministic reward structures. The first participant responded strongly to the anticipatory quality of randomized reward delivery, while the second found customization-based accumulation more aligned with their motivational preferences. This divergence supports the design decision to implement both a randomized mail system and a direct-purchase currency system in parallel, accommodating individual differences in reward sensitivity without requiring users to commit to a single progression logic.

These findings are preliminary and cannot be generalized beyond the two cases reported here. They are offered as an early indication of the design's motivational potential and as a basis for identifying specific refinements to be addressed in subsequent testing phases.

Conclusion

This study was developed in response to a specific and underexamined problem: the persistent gap between the theoretical promise of attention-support interventions for individuals with ADHD and their limited capacity to sustain user engagement over time. The central research question pursued was how a serious game might be structured to support focus initiation, task persistence, and longitudinal engagement for this population, not by circumventing the attentional and motivational characteristics of ADHD, but by building intervention logic around them.

The system developed in response to this question integrates a focus timer with a diegetic narrative framing, a journal-based task management structure, and a layered reward architecture operating across session, daily, and long-term timescales. Each of these components was derived from a convergence of existing literature on executive dysfunction and reward sensitivity in ADHD and insights generated through qualitative stakeholder interviews. Preliminary user testing with two participants diagnosed with ADHD produced early evidence that effort-contingent reward structures can meaningfully increase motivation and support task

initiation, particularly when the perceived value of rewards is sufficiently high. At the same time, testing surfaced two important limitations that constrain the current system's effectiveness: the attenuation of engagement over time consistent with a novelty effect, and the insufficiency of customization-based rewards alone to sustain a sense of cumulative achievement across extended use.

These findings carry specific implications for the next stage of development. The internal reward system must achieve a level of perceived salience comparable to the external reward conditions under which testing was conducted, which will require both expanding the item pool and strengthening the visibility of long-term progression. The divergence between participants in their responses to variable versus deterministic reward structures also suggests that future iterations should preserve both pathways rather than consolidating around a single reward logic.

More broadly, this study contributes to the literature on game-based digital therapeutics for ADHD by demonstrating that engagement architecture can be treated as a primary design problem rather than a secondary feature. The approach taken here, of grounding design decisions in the neuropsychological profile of the target population and iterating through qualitative user research, offers a methodological model for intervention design that prioritizes sustained participation as a precondition for therapeutic effect. Future work should extend this approach through larger-scale user testing, longitudinal engagement tracking, and clinical evaluation of whether the focus behaviors supported within the system transfer meaningfully to real-world academic and task contexts.

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